Quantum Traffic Gridlock

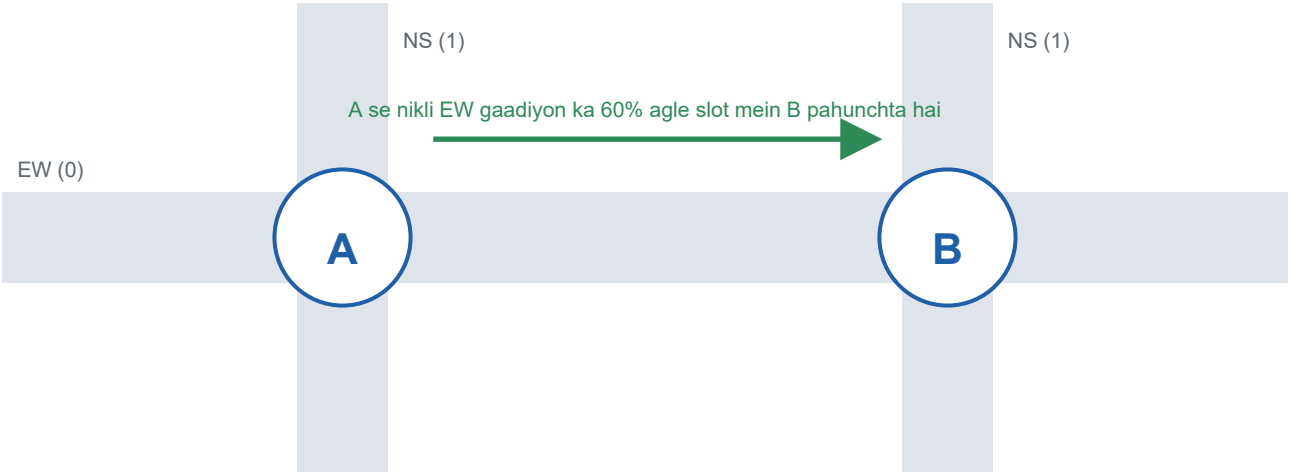
Quantum computer se traffic signal ka plan kaise banate hain — bilkul shuru se, aasaan bhasha mein. Kisi ko bhi samjhane ke liye.

> **Ek line mein:** Do jude hue traffic signals ka 4 slot ka plan banana tha. Humne traffic ke niyamon se quantum computer ke samajhne layak formula (QUBO) banaya, aur quantum algorithm (QAOA) ne 4096 koshishon mein zyaadatar achhe plan nikaale. Average cost **88** aaya, jabki sabse achha **72** aur bina soche (random) **151.25** hai.

1. Problem kya hai?

Ek sadak par do chauraahe hain, **A** aur **B**. A se nikli kuch gaadiyan aage jaakar B par pahunchti hain. Har chauraahe par ek waqt mein sirf ek taraf hara signal ho sakta hai:

- **NS** (North–South) — upar-neeche wali gaadiyan chalengi
- **EW** (East–West) — daayein-baayein wali gaadiyan chalengi

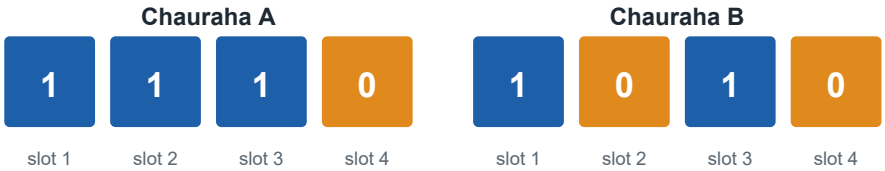


Chitra 1: do jude hue chauraahe (Scenario R mein link EW par hai)

Humein **4 time slots** (har slot 30 second) ka plan banana hai: har slot mein A par kaunsa signal hara ho, aur B par kaunsa. **Mushkil yeh hai** ki A ka faisla B ko prabhavit karta hai. A par EW hara kiya to woh gaadiyan agle slot mein B ki line badha deti hain. Isliye dono ko alag-alag nahi, saath mein sochna padta hai.

2. Plan ko kaise likhte hain? — 8 bits

Har slot ka faisla ek bit hai: **1 = NS hara**, **0 = EW hara**. 4 slot A ke + 4 slot B ke = **8 bits**.



Chitra 2: plan 1110 | 1010 (neela = NS, narangi = EW). Yahi Scenario R ka sabse achha plan hai.

8 bits se kul $2^8 = 256$ alag plan ban sakte hain. Inmein se sabse achha dhoondhna hai.

3. “Achha plan” kise kehte hain? — Cost

Organisers ne ek official formula (`scorer.py`) diya hai. **Jitni kam cost, utna achha plan.** Cost teen cheezon se banti hai:

Hissa	Matlab	Saza (Scenario R)
Delay	Har slot ke baad kitni gaadiyan abhi bhi line mein khadi hain	har gaadi = 1
Overflow	Line itni lambi ho gayi ki jagah (7 gaadi) se bahar nikal gayi	har extra gaadi = 4
Switch	Signal ek slot se agle slot mein badla	har badlav = 0.5

Har slot mein hara signal zyada se zyada **5 gaadiyan** nikaal sakta hai. Saari 256 plans valid hain, bas kuch mehenge hain.

4. Jab 256 hi plan hain to sab check kyon nahi karte?

Kar sakte hain, aur tulna (comparison) ke liye hum karte bhi hain. Lekin is competition ka maqsad yeh seekhna hai ki **quantum computer** aisi problem kaise solve karta hai. Asli shehar mein hazaaron chauraahe hote hain, tab sab plan check karna namumkin hai (2^{1000} se bhi zyada plan). Yahaan chhoti problem par tareeka test ho raha hai. PDF khud kehta hai: is size par quantum se fayde ki ummeed nahi hai.

5. Quantum computer ko problem kaise samjhate hain? — QUBO

Quantum algorithm sirf ek khaas tarah ka formula samajhta hai, jise **QUBO** kehte hain. Usmein sirf **akele bits** aur **do-do bits ke jode** ho sakte hain, teen ya zyada ek saath nahi. Asli traffic cost isse zyada tedhi hai: line kabhi negative nahi hoti, aur 5 se zyada gaadiyan ek slot mein nahi nikal sakti.

Humne kya kiya:

- **Tod diya:** kul cost = har line (A-NS, A-EW, B-NS, B-EW) ki cost + signal badalne ki cost. Yeh todna bilkul exact hai, official scorer se 256 × 40 plans par check kiya.
- **Signal badalna** pehle se jode wala formula hai, use exactly rakha.
- **Har line ki cost** ke liye sabse karib wala “jode-wala” formula banaya, jaise kisi tedhi line ko sabse achhi seedhi line se approximate karna (maths mein: order-2 Walsh projection).
- Yeh niyam har scenario par same chalta hai. Humne jawab pehle se dhoondhkar formula mein chhupaya **nahi** hai (yeh competition mein mana hai).

> **Kitna sahi hai?** Scenario R par is formula ka sabse achha plan bilkul asli best plan hi nikla (cost 72, galti 0%). 30 random scenarios mein average galti 0.9%, aur sabse kharab case mein 10.6%.

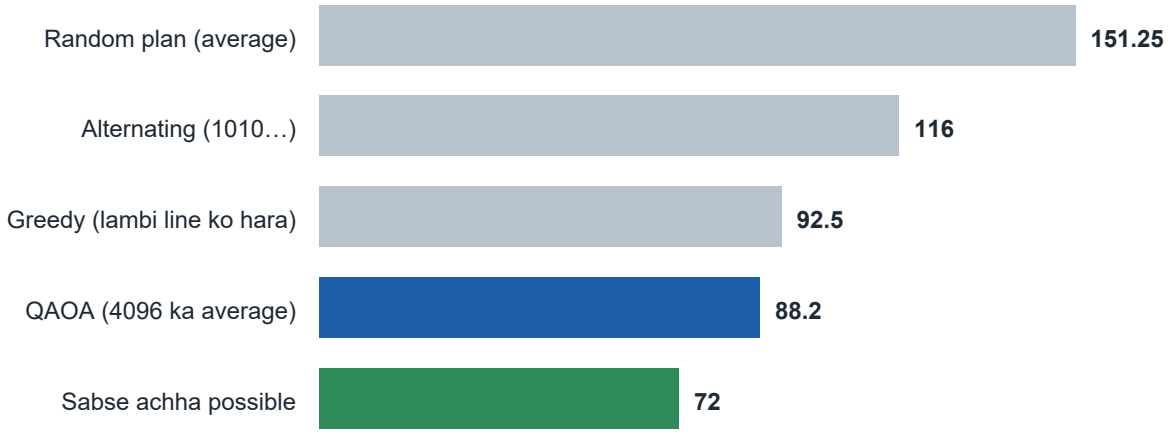
6. QAOA kya karta hai? — Quantum hissa

Socho 8 sikke hain, har sikka ek bit hai.

- **Shuru mein** saare sikke hawa mein ghoom rahe hain. Saare 256 plan ek saath, barabar chance (quantum superposition).
- **Fir do kaam baar-baar (4 baar, depth p = 4):** (a) *Cost step:* achhe plans ko thoda dhakka do; (b) *Mix step:* plans ko aapas mein milao taaki naye achhe plan bhi mil sakein.
- Kitna dhakka dena hai (angles), yeh ek normal computer **300 koshishon** ke andar seekhta hai. Pehle 1 layer seekhte hain, fir usi se 2, 3, 4 layers banate hain (INTERP tareeka).
- **Ant mein** circuit ko **4096 baar** chalakar dekhte hain ki kaunsa plan kitni baar aaya.

Kaamyabi kab maane? Jab achhe plans zyada baar niklein aur bure kam. Sirf ek baar best plan mil jaana kaafi nahi hai, poore 4096 nateejon ka average achha hona chahiye.

7. Nateeja — Scenario R



Chitra 3: cost (kam = behtar). QAOA ka average greedy se behtar hai, lekin best tak nahi pahuncha.

Metric	Value	Matlab
Lift	0.79 – 0.80	Random se best tak ka kitna faasla tay kiya (0 = random jaisa, 1 = perfect)
Elite enrichment	12.6×	Top 13 plans random ke mukable itne guna zyada baar aaye
Best plan kitni baar	8.3%	4096 mein se itne % baar sabse achha plan (72) aaya
5 seeds par	lagbhag same	Nateeja stable hai, kismat se nahi aaya

7b. Nateeja — Official Scenario H

Organisers ka doosra scenario H zyada bhaari traffic wala hai (capacity 4, link NS par). Code bina badle chalaya:

Tareeka	Cost (H)
Random plan (average)	300.3
Alternating	261
Greedy	244
QUBO formula ka best	236 (galti 0.9%)
QAOA (4096 ka average)	**242.3**
Sabse achha possible	234

H par QAOA ka average greedy se bhi behtar raha. Lift 0.87–0.88, aur top-13 plans random ke mukable ~17 guna zyada baar aaye.

8. Saath mein kya jaancha (baselines)

Tareeka	Kya karta hai	Cost (R)
Alternating	Har slot signal badlo (4 tarah ke)	116
Greedy	Har slot mein jo line lambi, use hara karo	92.5
Sab 256 check karo	Asli sabse achha jawab (sirf tulna ke liye)	72
Random	Bina soche plan ka average	151.25

QUBO ka best	Humare formula ka sabse achha plan	72
QAOA (raw)	Quantum ke 4096 nateejon ka average	88.2
Hybrid	QAOA + normal computer se chhota sudhaar (alag likha hai)	72

9. Imaandar baatein (limitations)

- Is chhoti problem par normal computer turant jawab de deta hai. Quantum se fayda dikhana maqsad nahi tha.
- Jode-wala formula hamesha perfect nahi hota. Ek random scenario mein yeh 10.6% peeche raha.
- Sab kuch quantum **simulator** par chala hai, asli quantum hardware par nahi. Hardware ka noise nahi jaancha.
- R aur H dono organisers ki official files hain. Judges ka unseen scenario S bhi isi code par bina badle chalega.

10. Shabdkosh (mushkil shabd)

Shabd	Aasaan matlab
Bit	0 ya 1 — yahaan: EW ya NS hara
Qubit	Quantum bit — ek saath 0 aur 1 dono ho sakta hai, jab tak dekho nahi
QUBO	Aisa cost formula jismein sirf akele bits aur do-do bits ke jode hon
QAOA	Quantum algorithm jo achhe jawabon ke chance badhata hai
Shot	Circuit ko ek baar chalakar ek jawab dekhna (humne 4096 shots liye)
Seed	Random numbers ka fixed number, taaki nateeja dobara same aaye
Baseline	Tulna ke liye simple tareeke (greedy, random, aadi)
Lift / Enrichment	QAOA random se kitna behtar hai, iske maap

Code chalane ke liye: `python run.py --scenario official/scenario_R.json --seed 0 --out outputs/R/seed_0`. Poori technical jaankari REPORT.md mein hai.